HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2024)

Student No.: 05 Name: _____ Marks: / 74

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. (a) Find the range of values of x which satisfy both $\frac{5x+7}{4} - 1 < 2x$ and $3x + 9 \geq 0$.

(b) Write down the least integer satisfying both inequalities in (a).

(4 marks) (DSE-2024 PI Q04)

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2. The table below shows the distribution of the numbers of keys owned by a group of housewives.

Number of keys	3	4	5	6	7	8
Number of housewives	10	9	4	3	4	k

If a housewife is randomly selected from the group, then the probability that she owns more than 6 keys is $\frac{5}{18}$.

(a) Find k.

(b) Write down the mean, the mode and the median of the distribution.

(5 marks) (DSE-2024 PI Q09)

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3. The stem-and-leaf diagram below shows the distribution of the numbers of hours spent on reading journals in a month by a group of researchers.

Stem (tens)	Leaf (units)
2	0 0 1 a a a 8 8 9 9
3	0 0 2 3 4 4 7 9
4	0 b

The mean of the distribution is 30 .

(a) Find a and b . (3 marks)

(b) Write down the least possible range of the distribution. (1 mark)

(c) Find the greatest possible inter-quartile range of the distribution. (3 marks)

(7 marks) (DSE-2024 PI Q11)

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4. Let $F(x) = (6x^2 + x + p)(qx^2 + rx - 10)$, where p, q and r are constants. The constant term of $F(x)$ is 40.

(a) Write down the value of p. (1 mark)

(b) When $F(x)$ is divided by $x + 1$, the remainder is -12. It is given that $x - 2$ is a factor of $F(x)$.

(i) Find q and r.

(ii) How many irrational roots does the equation $F(x) = 0$ have? Explain your answer. (7 marks)

(8 marks) (DSE-2024 PI Q14)

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5. The coordinates of the points Q and R are $(10, -1)$ and $(-4, -9)$ respectively.

(a) Let P be a moving point in the rectangular coordinate plane such that $PQ = PR$. Denote the locus of P by Γ.

(i) Describe the geometric relationship between Γ and QR.

(ii) Find the equation of Γ. (3 marks)

(b) Let C be the circle which passes through Q, R and the point $(4, 3)$.

(i) Find the equation of C.

(ii) The coordinates of the point U are $(10, 4)$. It is found that U lies outside C. UV and UW are the tangents to C at the points V and W respectively. Is the area of the circumcircle of $\triangle UVW$ greater than 100? Explain your answer. (5 marks)

(8 marks) (DSE-2024 PI Q17)

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6. (a) $PQRS$ is a thin quadrilateral metal sheet, where $PQ = 12\,\text{cm}$, $PS = 10\,\text{cm}$, $QR = 13\,\text{cm}$, $\angle QPS = 82^\circ$ and $\angle QRS = 65^\circ$. Find

(i) the length of QS,

(ii) $\angle RQS$. (4 marks)

(b) The metal sheet $PQRS$ described in (a) is now folded along QS (see Figure 2). It is given that the angle between the plane PQS and the plane QRS is 80°.

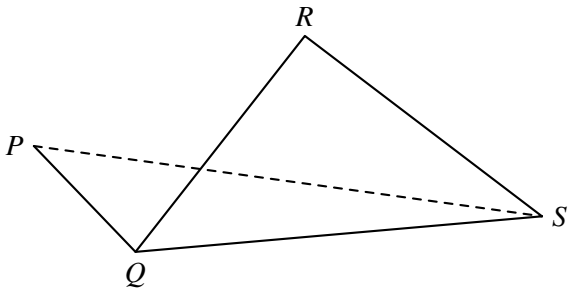


Figure 2

(i) Find the shortest distance from R to the plane PQS.

(ii) Let X be any point lying on the plane QRS. Someone claims that the distance between P and X exceeds $8\,\text{cm}$. Is the claim correct? Explain your answer. (4 marks)

(8 marks) (DSE-2024 PI Q18)

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7. Let $f(x) = 2x^2 + 4mx + 8x + 2m^2 + 8m + n$, where m and n are real constants such that $mn < 0$. Denote the vertex of the graph of $y = f(x)$ by P.

(a) Using the method of completing the square, express the coordinates of P in terms of m and n. (2 marks)

(b) Describe the geometric meaning represented by transforming $f(x)$ to $f\left(\frac{x}{5}\right) + 7$. (2 marks)

(c) Denote the vertex of the graph of $y = f\left(\frac{x}{5}\right) + 7$ by Q. Let (a_1, b_1) and (a_2, b_2) be the coordinates of P and Q respectively. It is given that a_1, $1 + n$, a_2 is an arithmetic sequence and b_1, $4 - m$, b_2 is a geometric sequence.

(i) Find the coordinates of P and Q.

(ii) The coordinates of the points R and S are $(3t + 27, t)$ and $(3t + 3, 2t - 3)$ respectively, where t is a real number. Is it possible that $PQRS$ is a rhombus? Explain your answer. (8 marks)

(12 marks) (DSE-2024 PI Q19)

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8. $\dfrac{(2a)^3}{(4a^{-5})^{-1}} =$

A. $2a^8$.

B. $32a^8$.

C. $\dfrac{2}{a^2}$.

D. $\dfrac{32}{a^2}$.

(2 marks) (DSE-2024 PII Q02)

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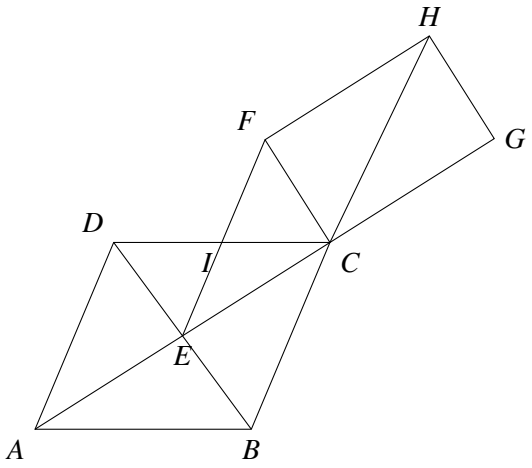
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9. In the figure, $ABCD$ is a rhombus. Denote the point of intersection of AC and BD by E. Let F be a point such that $BH//EF$ and $CFHG$ is a rectangle, where G and H are points lying on AC produced and BC produced respectively. Denote the point of intersection of CD and EF by I. Which of the following must be true?

I. $CI = FI$

II. $\angle ABE = \angle GCH$

III. $\triangle ADE \cong \triangle HCF$



A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q21)

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10. Find the constant k such that the straight lines $6x - 8y = 7k$ and $kx + 12y = 5$ do not intersect with each other.

A. -16

B. -9

C. 9

D. 16

(2 marks) (DSE-2024 PII Q26)

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11. Consider the following positive integers:

$$5 \quad 5 \quad 5 \quad 6 \quad 9 \quad 9 \quad 11 \quad 13 \quad m \quad n$$

Let p, q and r be the standard deviation, the mode and the median of the above positive integers respectively. If the mean of the above positive integers is 7, which of the following must be true?

I. $p > 3$

II. $q = 5$

III. $r < 7$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q30)

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12. $AF000000000BC_{16} =$

A. $175 \times 16^{11} + 188.$

B. $192 \times 16^{11} + 205.$

C. $175 \times 16^{12} + 188.$

D. $192 \times 16^{12} + 205.$

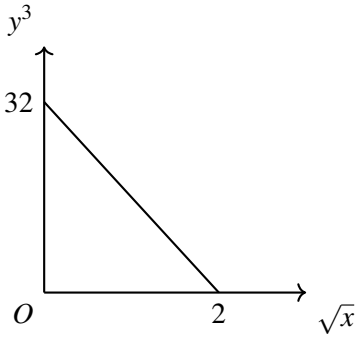
(2 marks) (DSE-2024 PII Q32)

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13. The graph in the figure shows the linear relation between y^3 and $\sqrt{x}$. If $x = 36$, then $y =$



A. -64.

B. -16.

C. -8.

D. -4.

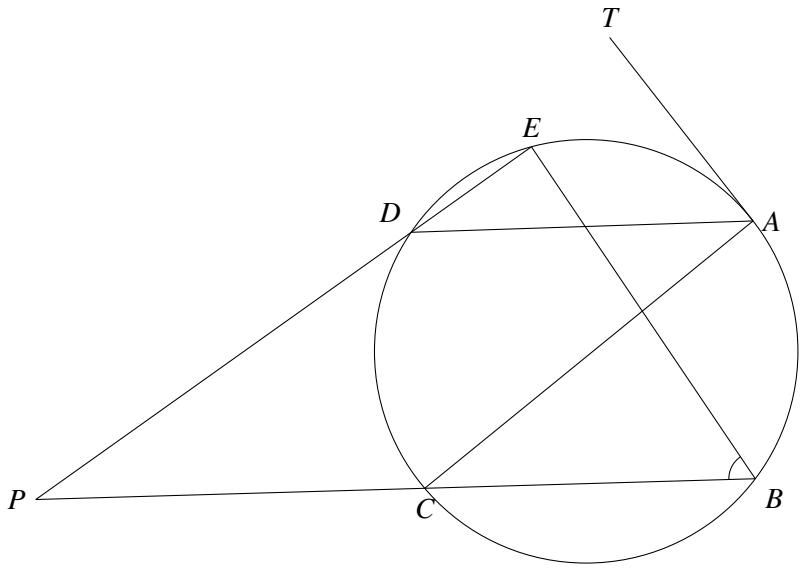
(2 marks) (DSE-2024 PII Q34)

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14. In the figure, TA is the tangent to the circle $ABCDE$ at the point A. BC produced and ED produced meet at the point P. If $\angle ACB = 43°$, $\angle DAT = 55°$ and $\angle CPD = 29°$, then $\angle CBE =$



A. 64°.

B. 69°.

C. 72°.

D. 78°.

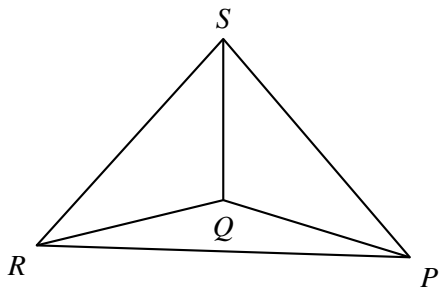
(2 marks) (DSE-2024 PII Q38)

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15. In the figure, the base PQR of the tetrahedron $PQRS$ lies on the horizontal ground. It is given that Q is vertically below S. If $\angle PQR = 90°$, $\angle QPS = 30°$ and $\angle QRS = 45°$, then $\cos\angle PRS =$



A. $\frac{1}{2}$.

B. $\frac{3}{4}$.

C. $\frac{\sqrt{2}}{4}$.

D. $\frac{\sqrt{3}}{6}$.

(2 marks) (DSE-2024 PII Q40)

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16. Let G, H, I and J be the centroid, the orthocentre, the in-centre and the circumcentre of $\triangle PQR$ respectively. If $\angle PQR = \angle PRQ = 22°$, which of the following are true?

I. G lies inside $\triangle PQR$.

II. H lies outside $\triangle PQR$.

III. I, J and Q are collinear.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q41)

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17. A queue is formed by 2 managers and 7 officers. If no managers are next to each other, how many different queues can be formed?

A. 80 640

B. 141 120

C. 282 240

D. 362 880

(2 marks) (DSE-2024 PII Q42)

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18. There are three questions in a test. The probabilities that a student answers the first question correctly, the second question correctly and the third question correctly are 0.6, 0.7 and 0.8 respectively. Find the probability that the student answers at least 1 question correctly in this test.

A. 0.024

B. 0.188

C. 0.812

D. 0.976

(2 marks) (DSE-2024 PII Q43)

I can finish _____ questions

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